

Not All Representations Are Subspaces

A Grassmannian atlas of causal structure in grokking

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2. DAS and the Grassmannian
3. Why linearity matters

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Part 1

What is a causal variable, and where does DAS look?

In this section

1. What is a causal variable?

When we say a neural network “uses” a concept, we mean there is a direction (or subspace) in its activations that, when edited, changes the output in a predictable way.

2. DAS and the Grassmannian

Distributed Alignment Search (DAS) finds these subspaces. Every DAS solution is a point on a specific mathematical object: the Grassmannian manifold. This is the hidden assumption.

3. Why linearity matters

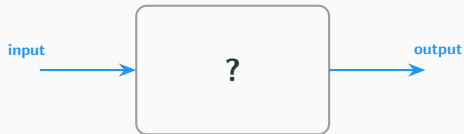
DAS assumes the causal variable is a linear subspace. This works for some variables but fails catastrophically for others — and high IIA scores cannot tell you which case you are in.

1. What is a causal variable?

The interpretability goal

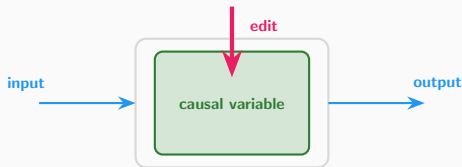
We want to understand *how* neural networks compute, not just *what* they compute.

Standard evaluation



Report accuracy, F1, etc.
Does not explain *why*

Causal abstraction



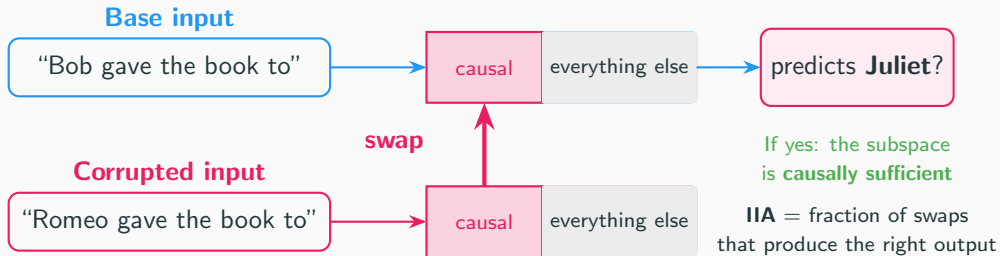
Find internal variables.
Edit them. Check if output changes.

If editing a specific direction in the network's internal representation changes the output predictably, that direction is a **causal variable** for the behavior.¹

¹Geiger et al., "Causal Abstraction for Faithful Model Interpretation," 2021.

Interchange intervention analysis (IIA)

The core technique: take two inputs, swap part of their internal representations, check if the output changes accordingly.²



²Geiger et al., "Inducing Causal Structure for Interpretable Neural Networks," ICML 2022.

2. DAS and the Grassmannian

What DAS actually does

DAS learns a rotation matrix $Q \in \mathbb{R}^{d \times k}$ that identifies a k -dimensional subspace of the d -dimensional activation space.^a

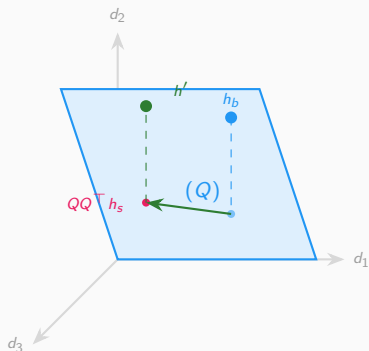
The interchange intervention projects out the base's component and replaces it with the source's:

$$h' = h_b - QQ^\top h_b + QQ^\top h_s$$

This is a **linear** operation. The subspace $QQ^\top$ is an orthogonal projection onto a *flat plane* in activation space.

The set of all k -dimensional planes in $\mathbb{R}^d$ is the **Grassmannian manifold** $\text{Gr}(k, d)$.

Every DAS solution is a point on $\text{Gr}(k, d)$.



DAS projects onto a **flat plane**.

What if the variable lives on a **curve**?

The Grassmannian manifold $\text{Gr}(k, d)$

The Grassmannian $\text{Gr}(k, d)$ is the set of all k -dimensional linear subspaces of $\mathbb{R}^d$.

Think of it as a “space of planes.” Every DAS solution picks one plane from this space. The distance between two planes is measured by their **principal angles**:

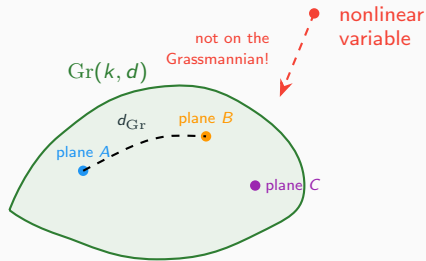
$$d_{\text{Gr}}([Q_1], [Q_2]) = (\sum_i \theta_i^2)^{1/2}$$

When we ask “does DAS work?” we are really asking:

Does the true causal variable live on $\text{Gr}(k, d)$?

If yes: DAS finds a meaningful point.

If no: DAS still returns a point, but it is an **artifact**.



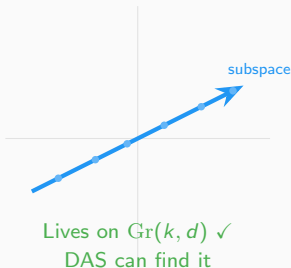
A nonlinear causal variable (circle, sphere)

does not live on $\text{Gr}(k, d)$ at any k .

3. Why linearity matters

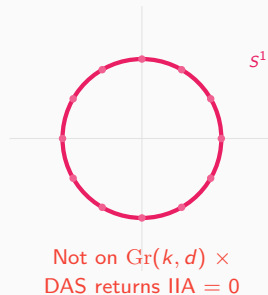
Linear vs. nonlinear causal variables

Linear causal variable



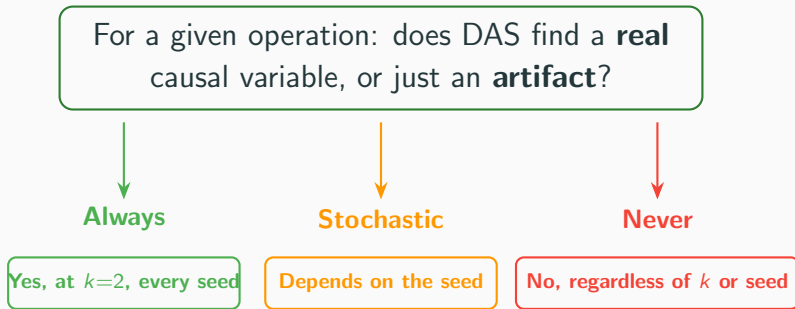
Examples: IOI indirect object (partially),
multiplication mod p after grokking

Nonlinear causal variable



Examples: addition mod p after grokking
(Fourier representation encodes numbers on a
circle)

The central question



We test fourteen modular arithmetic operations and find a **sharp, three-class partition**.

Why subspaces matter for circuits

Mechanistic interpretability asks two questions about neural network circuits:

Q1: Which components?

Which attention heads and MLPs matter for a behavior? Which edges carry the signal?

Methods: activation patching, attribution patching, EAP.

Q2: What do they communicate?

When two components are connected, *what* flows between them? Which subspaces of the residual stream carry the signal?

Methods: DAS, probing, SAEs.



Q1: *this edge matters*



Q2: *factor f_i carries the signal*

Our work: whether Q2 has a **linear** answer depends on the operation.

Part 2

The atlas: fourteen operations, three classes

4. Fourteen operations, three classes

We train on 14 modular arithmetic operations and find a sharp partition into three classes based on whether DAS finds a genuine causal variable.

5. Linear DAS fails completely

On grokked addition, DAS returns $\text{IIA} = 0$ at every k . The Fourier representation lives on S^1 , not on $\text{Gr}(k, d)$.

6. Stochastic grokking

Power and composite addition grok stochastically: same operation, same hyperparameters, opposite outcomes from different random seeds.

4. Fourteen operations, three classes

Experimental setup

We train single-layer transformers on modular arithmetic:

Input: $(a, b, =)$ Output: $f(a, b) \bmod p$

Model: $d_{\text{model}} = 128$, 4 heads, standard grokking setup (30% train, AdamW, weight decay 1.0).

For each trained model, we:

1. Cache residual-stream activations
2. Train DAS at $k \in \{2, 4, 6, \dots, 32\}$
3. Measure IIA, equivariance, circle geometry
4. Train a structured VAE

14 operations spanning group actions, polynomials, piecewise, and non-algebraic maps.

Group actions (5)

$$ab \bmod p$$

$$(a+b) \bmod 91$$

$$(a-b) \bmod p$$

$$a/b \bmod p$$

$$a \oplus b \bmod p$$

Polynomials (6)

$$a^2+b^2, a^3+b^3, a^3$$

$$a^2, a^2+b, 2a+3b+5$$

Piecewise (2)

$$\max(a, b), |a-b|$$

Non-algebraic (1)

$$a^b \bmod p$$

What is grokking?

Grokking is a training phenomenon where models memorize the training set long before they generalize.^a

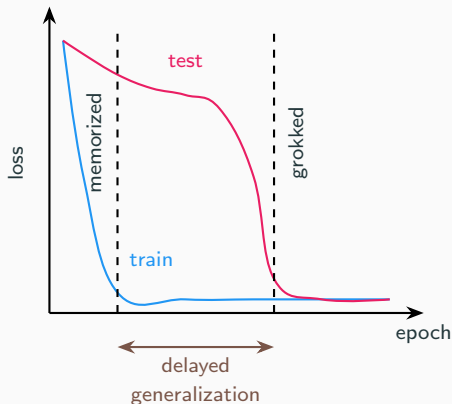
For modular arithmetic with weight decay:

- Train loss drops to zero quickly (memorization)
- Test loss stays high for thousands of epochs
- Then suddenly drops (generalization)

The internal representation changes during grokking: from a lookup table (memorization) to a structured algorithm (Fourier representation).

Prior work showed grokked models use Fourier features^b:

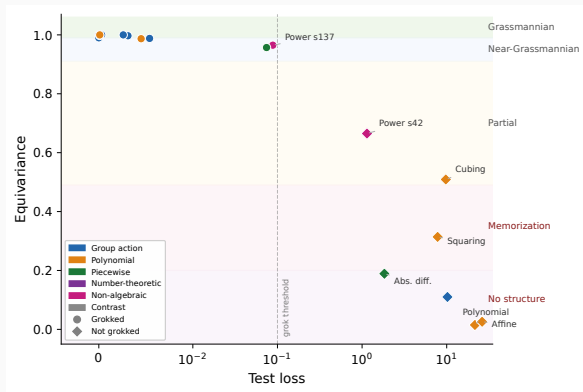
$$a \mapsto (\cos 2\pi fa/p, \sin 2\pi fa/p)$$



The atlas: three sharp classes

Operation	Grok	Loss	IIA($k=2$)	k^*	Equiv.	Class
Multiplication	✓	0.000	1.000	2	0.995	Always
Subtraction	✓	0.000	1.000	2	1.000	Always
Division	✓	0.000	1.000	2	1.000	Always
Bitwise XOR	✓	0.003	1.000	2	1.000	Always
Cubic sum	✓	0.000	1.000	2	1.000	Always
Sum of squares	✓	0.005	1.000	2	0.987	Always
Max	✓	0.074	0.960	2	0.957	Always
Power (s137)	✓	0.088	0.940	4	0.965	Stochastic
Power (s42)	×	1.143	0.980	2	0.665	Stochastic
Comp. add. (orig)	✓	0.002	1.000	2	0.998	Stochastic
Comp. add. (s42)	×	10.21	0.800	>32	0.110	Stochastic
Cubing	×	9.765	0.857	4	0.509	Never
Squaring	×	7.808	0.857	4	0.314	Never
Abs. difference	×	1.835	0.800	>32	0.189	Never
Polynomial	×	21.55	0.720	>32	0.015	Never
Affine	×	26.21	0.780	>32	0.026	Never

Loss vs. equivariance: grokking is the boundary



Each point is one (operation, seed) pair.

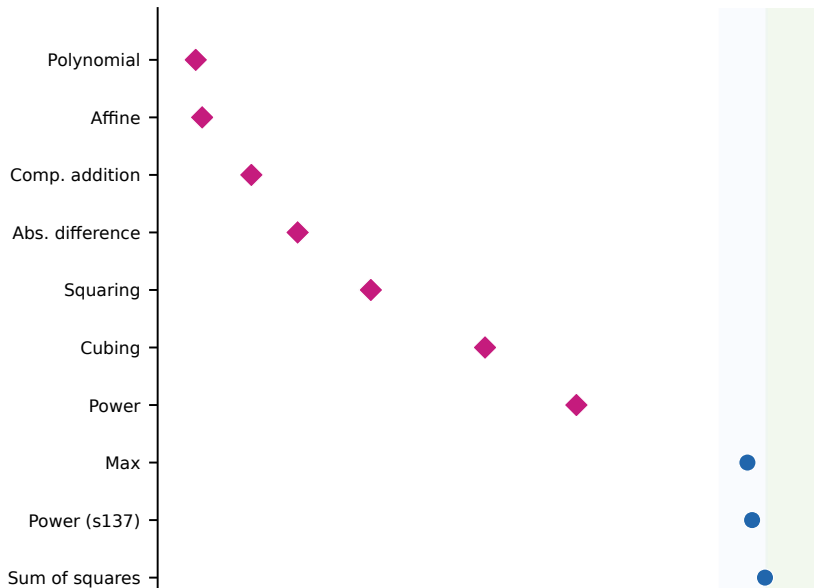
Blue: grokked models (test loss < 0.1).

Pink: non-grokked models.

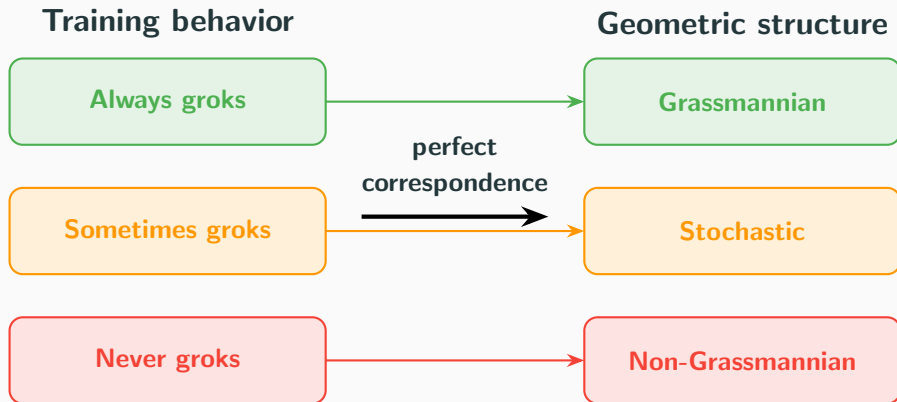
The boundary is sharp: grokking $\Leftrightarrow$ high equivariance $\Leftrightarrow$ Grassmannian causal variable.

No operation sits in the middle. The three-class partition is not a continuum.

Equivariance across all operations and seeds



The finding: grokking predicts geometry



Grokking is the mechanism: when generalization happens, the representation becomes a linear subspace on $\text{Gr}(k, d)$. When it doesn't, the variable stays nonlinear.

Three classes, one mechanism

Always Grassmannian

7 operations

- All grokked
- IIA = 1.0 at $k = 2$
- Equivariance > 95%
- Causal variable is a linear subspace



Stochastic

2 operations (4 seeds)

- Same operation
- Same hyperparameters
- Different random seed
- Opposite outcomes!
- Grokking is the switch



Never Grassmannian

5 operations

- None grokked
- IIA high but via lookup
- Equivariance < 52%
- Causal variable is nonlinear or absent



5. Linear DAS fails completely

Linear DAS returns zero IIA on grokked modular addition

Standard addition mod 113 (prime) groks perfectly (test loss $< 10^{-7}$). The model has learned the correct algorithm.

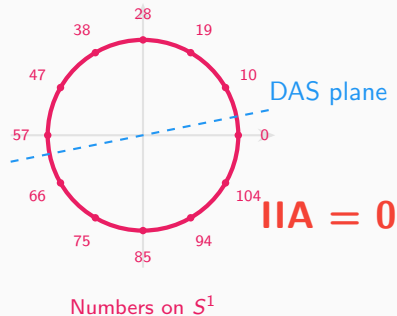
But DAS returns $\text{IIA} = 0.0$ at **every** subspace dimension $k \in \{2, 4, 8, 16\}$.

This is not an optimization failure. The model encodes numbers using Fourier features:

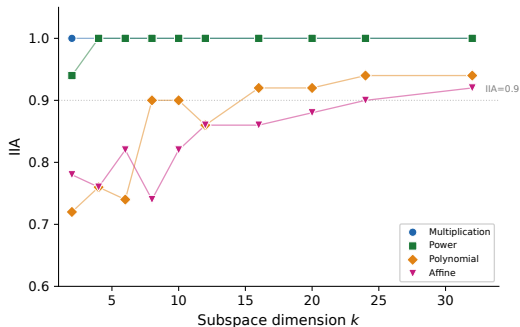
$$a \mapsto \left(\cos \frac{2\pi fa}{p}, \sin \frac{2\pi fa}{p} \right)$$

This representation lives on a **circle** S^1 in $\mathbb{R}^d$, not a linear subspace.

DAS searches $\text{Gr}(k, d)$ for a flat plane that captures a curved manifold — a **category error**, not an optimization failure.



k -sweep profiles distinguish the three classes



Always Grassmannian: IIA saturates at 1.0 by $k = 2$. The causal variable is intrinsically 2-dimensional.

Stochastic: power jumps to 1.0 at $k = 4$, depending on seed.

Never Grassmannian: IIA rises gradually and may not reach 1.0 even at $k = 32$. More dimensions help, but the fit is never clean.

The k -sweep profile is a **diagnostic**: if IIA keeps rising past $k = 4$, the causal variable is probably not Grassmannian.

6. Stochastic grokking

Stochastic grokking: same operation, opposite outcomes

The stochastic class is the **strongest evidence**:

Power ($a^b \bmod 113$):

- Seed 137: grokked. Equiv = 96.5%
- Seed 42: not grokked. Equiv = 66.5%

Composite addition ($(a + b) \bmod 91$):

- Original seed: grokked. Equiv = 99.8%
- Seed 42: not grokked. Equiv = 11.0%

Same operation. Same architecture. Same hyperparameters. **Opposite outcomes from random initialization alone.**

Grassmannian variables appear if and only if the model generalizes.

Same operation
Same architecture
Same hyperparams
Different seed

Power: seed 137

Grokked
Loss = 0.088
Equiv = 96.5%

$\neq$

Power: seed 42

Not grokked
Loss = 1.143
Equiv = 66.5%

Equivariance: the geometric diagnostic

For operations with a group action, the DAS subspace should respect the algebraic symmetry.

Test: shift input $a \rightarrow a + 1 \bmod p$. Does the DAS projection rotate by $2\pi/p$?

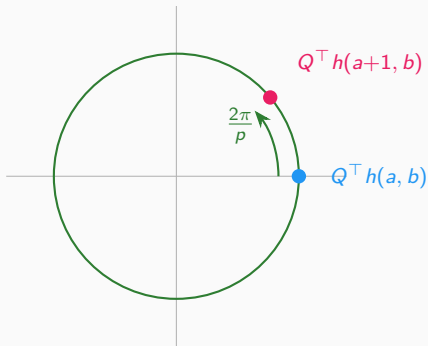
Random subspaces: equivariance $< 1\%$.

Always Grassmannian: equivariance $> 95\%$.

Never Grassmannian: equivariance $< 52\%$.

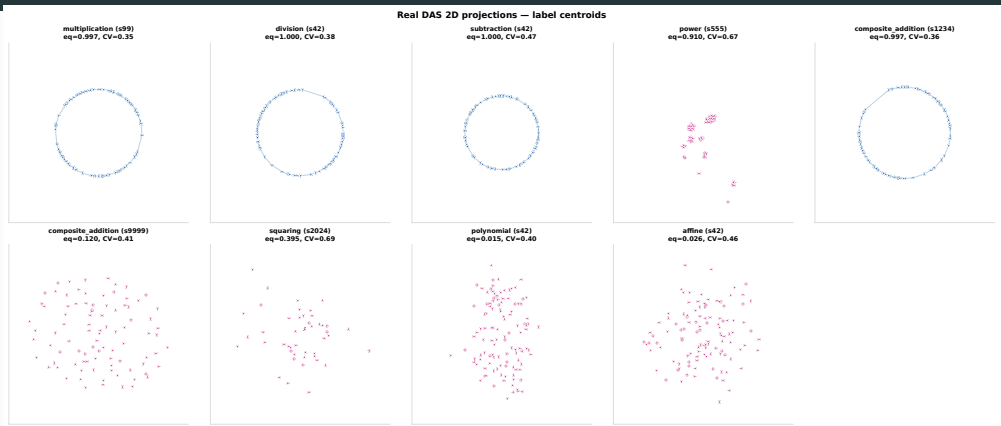
Equivariance distinguishes:

- Genuine structure from memorization
- Causal variables from artifacts
- What IIA alone cannot tell apart



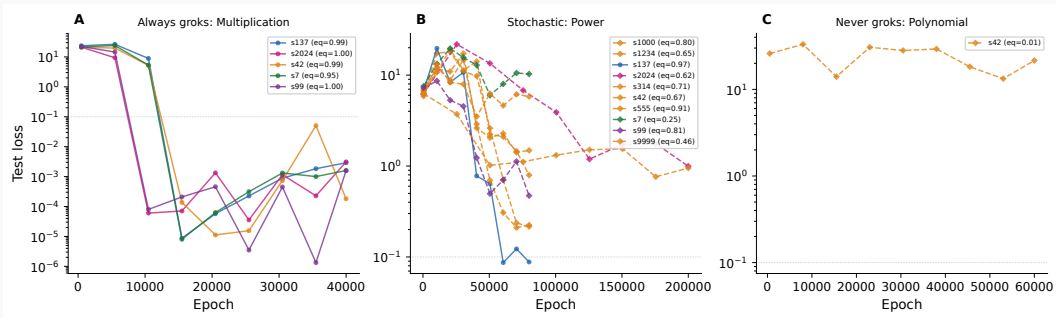
**If the variable is Grassmannian,
shifting input = rotating projection**

Circle geometry: DAS projections reveal structure



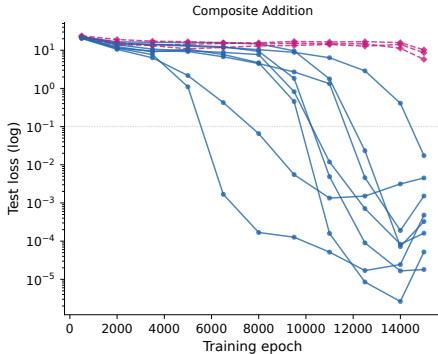
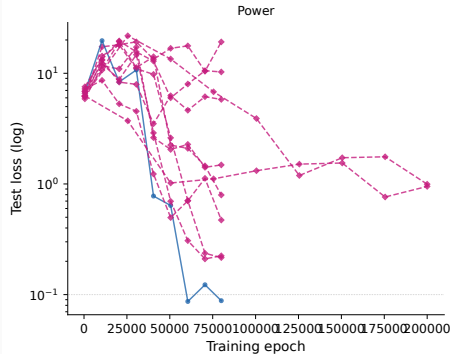
Blue: grokked operations form clean circles (label centroids on S^1). **Pink:** non-grokked operations show random scatter. Each panel shows the $k=2$ DAS projection with per-label centroids overlaid.

Training trajectories: the three classes in action



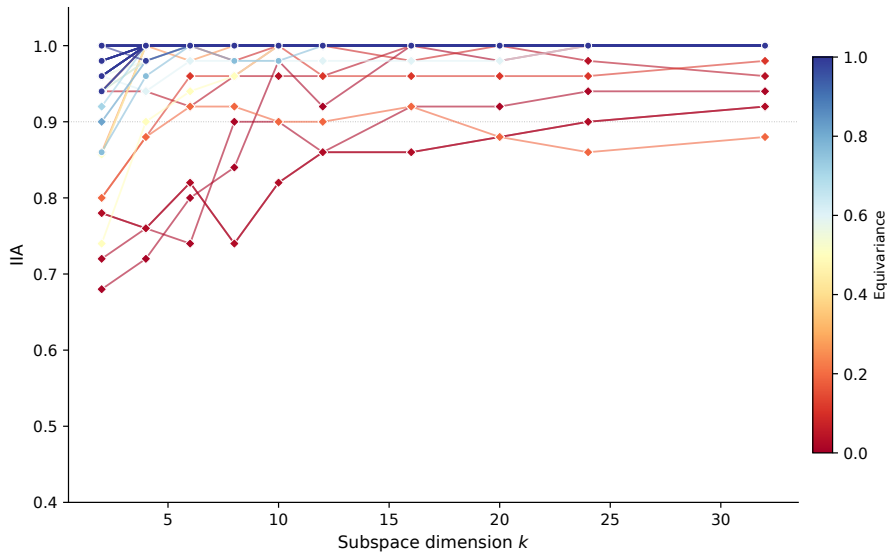
A: Multiplication always groks (all seeds, loss $\rightarrow 0$). **B:** Power is stochastic (some seeds grok, others plateau). **C:** Polynomial never groks (loss stays high across all seeds). Equivariance values in parentheses track grokking perfectly.

Stochastic grokking: power and composite addition



Multi-seed training for the two stochastic operations. Solid lines with circles = grokked (high equivariance). Dashed lines with diamonds = not grokked (low equivariance). Same operation, same hyperparameters, opposite outcomes from random initialization.

Full k -sweep across all operations



Part 3

The vacuity problem: when high IIA means nothing

7. Memorization artifacts

High IIA does not imply genuine structure.
Squaring achieves $\text{IIA} = 1.0$ via
memorization, exposed by low equivariance.

8. Nonlinear DAS is vacuous

NL-DAS achieves $\text{IIA} = 1.0$ on IOI at $k=1$ by
learning a lookup table. The diversity ratio
 $\rho \approx 0$ reveals the failure.

9. Intervention faithfulness

We introduce four complementary metrics —
diversity ratio, reconstruction MSE, KL
divergence, normalized logit diff — that
distinguish genuine interventions from
artifacts.

7. Memorization artifacts

Memorization produces high IIA

Squaring ($a^2 \bmod 113$) achieves $\text{IIA} = 0.86$ at $k = 2$ and $\text{IIA} = 1.0$ by $k = 4$.

But the model **never grokked** (test loss = 7.8). It memorized the training set.

Equivariance exposes the artifact:

- Squaring: 31.4% (vs. $> 95\%$ for genuine)
- Cubing: 50.9%
- Random subspace: $< 1\%$

High IIA with low- k saturation is compatible with a **lookup table**. IIA alone cannot tell the difference.

IIA alone
cannot tell
these apart!

Genuine (multiplication)

$\text{IIA} = 1.0$ at $k = 2$
Equiv = 99.5%

✓ Structured causal variable

Artifact (squaring)

$\text{IIA} = 1.0$ at $k = 4$
Equiv = 31.4%

× Lookup table

8. Nonlinear DAS is vacuous

The natural response: add nonlinearity

DAS assumes linearity and fails on nonlinear variables. Natural fix: prepend an MLP featurizer.

NL-DAS: learn encoder f and decoder g , then apply linear DAS in the transformed space:

$$h' = g(f(h_b) - QQ^\top f(h_b) + QQ^\top f(h_s))$$

Result on IOI^a (GPT-2):

- DAS at $k = 1$: IIA = 0.193
- NL-DAS at $k = 1$: IIA = **1.000**

This looks like a dramatic improvement.

It is not.

DAS (linear)

$$h \xrightarrow{QQ^\top} h'$$

NL-DAS (nonlinear)

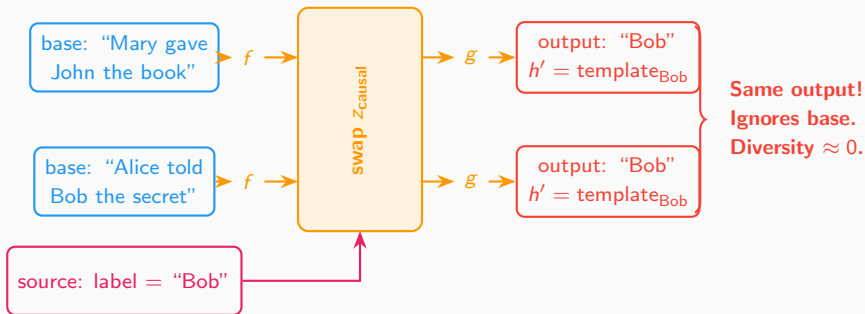
$$h \xrightarrow{f} z \xrightarrow{QQ^\top} z' \\ \xrightarrow{g} h'$$

**f and g trained only
on interchange loss**

No reconstruction constraint!
 $g \circ f \neq \text{identity}$

The lookup-table failure mode

What NL-DAS actually learns



The decoder produces a **canonical activation for each class**, regardless of base input.

This achieves $\text{IIA} = 1.0$ but is a **lookup table**, not a causal variable.

NL-DAS vs. DAS vs. structured VAE

Task	k	DAS	NL-DAS	VAE	Assessment
IOI	1	0.193	1.000	0.232	NL-DAS vacuous
IOI	2	0.297	1.000	0.719	NL-DAS vacuous
IOI	4	0.556	1.000	0.978	VAE genuine
Addition	1	0.006	0.194	0.000	Both honest
Addition	4	0.289	0.883	0.089	NL-DAS suspect
Mult.	4	0.344	0.961	0.000	NL-DAS suspect
Squaring	*	0.000	0.011	0.000	No structure

Red bold: suspiciously high IIA. **Green bold:** validated by diversity ratio.

IOI rows use GPT-2; grokking rows use single-layer models (same as atlas but with NL-DAS/VAE applied). NL-DAS achieves $\text{IIA} \approx 1$ on IOI at $k = 1$ — perfect interchange from a **single dimension**. Diversity ratio ≈ 0 confirms it is a lookup table.

9. Intervention faithfulness

The diversity ratio: detecting lookup tables

For each source label y_s , compute the standard deviation of intervened activations h' across all base inputs:

$$\rho = \frac{\mathbb{E}_{y_s} [\text{std}(h'_{y_s})]}{\mathbb{E}_{y_s} [\text{std}(h_{y_s})]}$$

$\rho \approx 0$: all intervened activations for the same source label collapse to a single point. **Lookup table.**

$\rho \approx 1$: the intervention preserves natural variation within each class. **Genuine intervention.**

This is the key metric that exposes NL-DAS: it achieves $\text{IIA} = 1.0$ but $\rho \approx 0$.

Genuine ($\rho \approx 1$)



Lookup table ($\rho \approx 0$)



Same source label y_s ,
different base inputs

Four complementary faithfulness metrics

Diversity ratio ρ

Does the intervention preserve variation within each class?

$\rho \approx 0$: lookup table

$\rho \approx 1$: genuine

Reconstruction MSE

Can the decoder reconstruct the original activation?

High MSE + high IIA:
decoder is “hallucinating”

KL / JS divergence

How much does the full output distribution change?

Lower = more faithful.

Binary IIA hides this.

Normalized logit diff

Does the intervention reproduce the clean model's confidence?

1.0 = exact match

$\gg 1$ or $\ll 0$ = distortion

Why this matters

Binary IIA treats a 51% probability shift the same as a 99% shift.

These continuous metrics distinguish among methods that all achieve $\text{IIA} \approx 1.0$.

Report at least one faithfulness metric alongside IIA.

Part 4

The fix: structured VAE and cross-task transfer

In this section

10. Structured VAE

A VAE with causal/nuisance split and ELBO prevents the lookup-table failure. Matches DAS where it works, extends to nonlinear.

10b. Structured pi-SAE (ours)

Adding L1 sparsity to the label-conditional prior transforms pi-VAE from $\text{IIA} = 0$ to $\text{IIA} = 1$ on grokked operations. Reveals true intrinsic dimension. CRT composite moduli confirm genuinely 2D causal variables.

10c. NLP tasks

Structured pi-SAE on GPT-2 language tasks (SVA, greater-than, gender bias, capitals,

12. IOI is not Grassmannian

Sheaf consistency shows IOI's variable is locally linear but globally curved.

From subspaces to factors

Factorized DAS finds interpretable factor directions within the subspace.

Weight space vs. activation space

Factors decompose how the model computes, not just what it outputs.

10. Structured VAE

Structured VAE for nonlinear causal variables

The structured VAE partitions the latent space into:

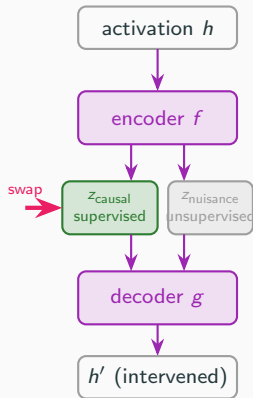
- z_{causal} : supervised with the operation's label
- z_{nuisance} : unsupervised, absorbs everything else

The loss combines three terms:

$$\mathcal{L} = \underbrace{-\mathbb{E}_q[\log p(x | y, z)]}_{\text{reconstruction}} + \beta \underbrace{D_{\text{KL}}}_{\text{regularization}} + \alpha \underbrace{\mathcal{L}_{\text{CE}}}_{\text{supervision}}$$

The ELBO prevents the lookup-table failure:

1. Reconstruction forces faithful decoding
2. KL prevents latent collapse
3. Causal/nuisance split preserves base info



Why the VAE is not vacuous: three constraints

NL-DAS (vacuous)

- × No reconstruction loss
- × No KL regularization
- × No causal/nuisance split
- × Trained only on IIA
- Can learn lookup table
- Diversity $\rho \approx 0$

VS.

Structured VAE (genuine)

- ✓ Reconstruction → faithful decoding
- ✓ KL → prevents latent collapse
- ✓ Causal/nuisance split
- ✓ ELBO + supervision
- Cannot learn lookup table
- Diversity $\rho \approx 1$

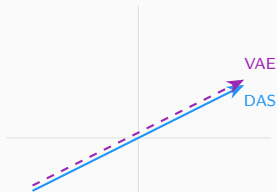
Controls confirm the VAE is not cheating

Random-label VAE: $\text{IIA} \leq 0.02$ | Reconstruction-only ($\alpha = 0$): $\text{IIA} \leq 0.005$ | Untrained: $\text{IIA} = 0.000$
Plain VAE (no causal/nuisance split): $\text{IIA} \leq 0.08$ — Correct supervision AND structure are both required

VAE matches DAS where DAS works, extends where it fails

Always Grassmannian operations:

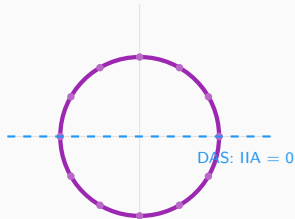
VAE reduces to an effective rotation.



Both find the same
linear subspace

Never Grassmannian operations:

VAE's nonlinear encoder captures curved structure.



VAE captures nonlinear
causal structure

For grokked addition and multiplication, both DAS and VAE achieve $\text{IIA} = 1.000$ and equivariance $> 99\%$.

The VAE is a **strict generalization** of DAS: when the variable is linear, $\text{VAE} = \text{DAS}$.

10b. Structured pi- π -SAE: sparse nonlinear causal discovery

From pi-VAE to Structured pi-SAE: sparsity + structure

pi-VAE (Zhou & Wei, 2020)

Label-conditional prior:

$$p(z \mid y) = \mathcal{N}(\mu(y), \sigma^2(y))$$

Instead of the standard $\mathcal{N}(0, I)$ prior, each label gets its own mean and variance. Forces the latent space to organize by label.

Published method. Works well on IOI.

But on grokked modular arithmetic:

IIA = 0 across all k .

Structured pi-SAE (ours)

pi-VAE + **causal/nuisance split** + **L1 sparsity**:

$$\mathcal{L} = \mathcal{L}_{\text{pi-VAE}} + \lambda \|z_c\|_1, \quad z = (z_c, z_n)$$

Explicit causal/nuisance partition forces the model to separate task-relevant from task-irrelevant variation. L1 sparsity on z_c prevents memorization.

Our modification. Results:

	pi-VAE	Str. pi-SAE
Addition	0.000	1.000
Multiplication	0.000	1.000
Quantization	0.000	1.000

Intrinsic dimension: $k=1$ reveals the true structure

A circle is topologically **1-dimensional** — parameterized by a single angle θ .

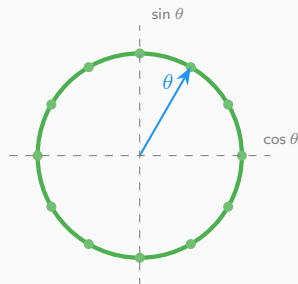
Linear DAS needs $k=2$ because it embeds the circle as $(\cos \theta, \sin \theta)$ on a flat plane.

A nonlinear method can learn $z = \theta$ directly and decode back to the circle. **Structured pi-SAE at $k=1$:**

	DAS	NL-DAS	Structured pi-SAE
Addition	0.006	0.194	0.917
Multiplication	0.022	0.217	0.011
Quartic sum	0.011	0.244	0.106
IOI	0.185	1.000	0.988

Addition's circle admits a smooth 1D

parameterization. Multiplication's discrete log



Linear: needs $k=2$
Nonlinear: needs $k=1$

Linear DAS **overestimates**
intrinsic dimension.

Structured pi-SAE results across all operations

Operation	Grok	Acc	IIA at $k=2$			IIA at $k=1$		
			DAS	NL-DAS	Str. pi-SAE	DAS	NL-DAS	Str. pi-SAE
Addition	✓	1.00	0.06	0.49	1.00	0.01	0.19	0.92
Multiplication	✓	1.00	0.08	0.72	1.00	0.02	0.22	0.01
Quartic sum	✓	1.00	0.03	0.79	1.00	0.01	0.24	0.11
IOI (GPT-2)	—	—	0.30	1.00	0.98	0.19	1.00	0.99
Mixed product	✗	0.15	0.02	0.60	0.04	0.01	0.21	0.02
Symmetric power	✗	0.21	0.02	0.63	0.02	0.01	0.26	0.02
Double add mult	✗	0.10	0.00	0.77	0.02	0.00	0.22	0.02
Squaring	✗	0.01	0.00	0.01	0.00	0.00	0.01	0.00

Green bold: genuine causal variable found. **Red:** NL-DAS vacuity — high IIA despite no learned structure.

Structured pi-SAE achieves $\text{IIA} = 1.0$ on all grokked operations and correctly returns ≈ 0 on all

Ablation: both sparsity and structure are required

IIA at $k=2$ across the 2×2 ablation: $\{\text{structured prior, plain prior}\} \times \{\text{VAE, SAE}\}$.

Operation	Plain prior $\mathcal{N}(0, I)$		Structured prior $p(z y)$	
	VAE	SAE	pi-VAE	Str. pi-SAE
Addition	0.02	1.00	0.00	1.00
Multiplication	0.00	0.72	0.00	1.00
Quartic sum	0.02	0.32	0.00	1.00
IOI	0.56	0.83	0.95	0.98
Mixed product	0.03	0.01	0.01	0.04
Squaring	0.00	0.00	0.00	0.00

Plain SAE works on addition but degrades on multiplication and quartic sum.

Structured pi-SAE is uniformly best.
Structured prior + sparsity
= robust across all tasks.

Structured pi-SAE exposes NL-DAS vacuity

The problem:

NL-DAS achieves $IIA = 0.6\text{--}0.8$ on operations that **never learned** the task (test accuracy $< 20\%$).

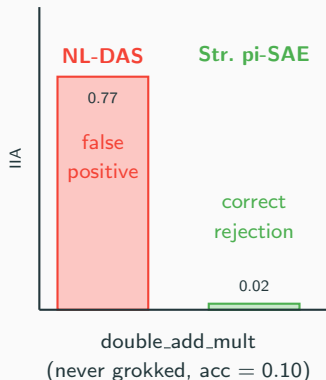
These are false positives. The encoder-decoder learned a lookup table, not a causal variable.

Structured pi-SAE correctly rejects:

On the same operations, Structured pi-SAE returns $IIA \leq 0.04$. The reconstruction constraint prevents hallucination.

Reconstruction MSE confirms:

Grokked Not grokked



CRT composite modulus: the first task needing $k > 1$

Chinese Remainder Theorem.

For composite modulus $n = p \times q$:

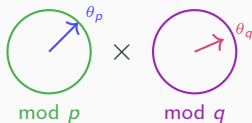
$$\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/q\mathbb{Z}$$

The model learns **two independent circles** — one mod p , one mod q .

Intrinsic dimension = **2** (two angles θ_p, θ_q).

Linear DAS needs $k \geq 4$ (cos / sin for each).

Structured pi-SAE captures both at $k=2$.



Task	k	DAS	NL-D	Str. pi-SAE	
add 113	1	0.01	0.19	0.92	✓
(prime)	2	0.06	0.49	1.00	
add mod 77	1	0.02	0.19	0.06	✓
(7 × 11)	2	0.07	0.59	0.99	
add mod 91	1	0.02	0.24	0.00	✓
(7 × 13)	2	0.16	0.70	0.98	
add mod 35	2	0.08	0.51	0.21	×
(5 × 7)	4	0.21	0.95	0.21	

Green: genuine. **Red:** NL-DAS false positive.

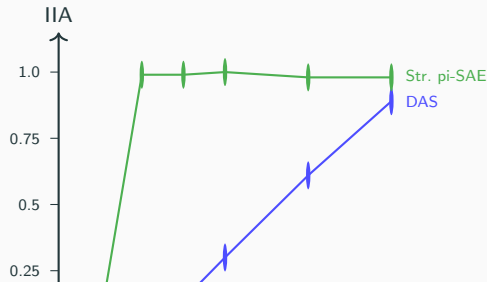
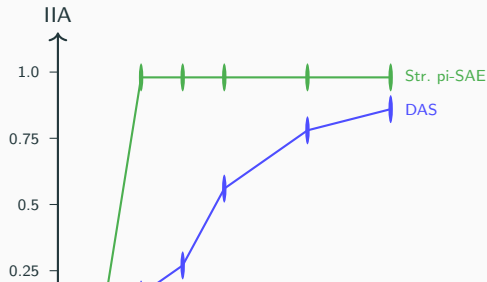
$k=1$ fails on CRT tasks $\Rightarrow$ intrinsic dim is truly 2.

Structured pi-SAE $k=2$ beats DAS $k=8$ (0.86–0.89).

CRT k -sweep: linear DAS needs $4\times$ more dimensions

Addition mod 91 ($= 7 \times 13$). Full k -sweep from 1 to 8:

Method	IIA at each k					
	$k=1$	$k=2$	$k=3$	$k=4$	$k=6$	$k=8$
DAS (linear)	0.02	0.16	0.27	0.56	0.78	0.86
NL-DAS	0.24	0.70	0.94	0.92	0.92	0.88
Str. pi-SAE (ours)	0.00	0.98	0.98	0.98	0.98	0.98



Intrinsic dimension hierarchy: prime vs. composite

The minimum k at which Structured pi-SAE achieves $\text{IIA} \geq 0.9$ reveals the **true intrinsic dimension** of the causal variable:

Operation	Structure	Min k	Manifold
add mod 113	prime	1	S^1
add mod 77	7×11	2	$S^1 \times S^1$
add mod 91	7×13	2	$S^1 \times S^1$
IOI (GPT-2)	NLP	1	$S^1?$

Prediction: addition mod $p_1 \cdots p_n$ needs $k = n$.

Testable: mod 2310 = $2 \times 3 \times 5 \times 7 \times 11$ would need $k=5$.

Linear DAS always overestimates:

Linear k Nonlinear k

$k = 1$: **one circle**



mod p (prime)

$k = 2$: **torus** $S^1 \times S^1$



mod p

mod q

Nonlinear methods reveal the manifold's true topology, not its linear embedding dimension.

10c. Beyond grokking: NLP tasks on GPT-2

Structured pi-SAE on GPT-2 language tasks

Beyond toy grokking models: apply Structured pi-SAE to **real NLP tasks** in GPT-2-small at layer 10.

Using MIB counterfactual pairs (Hanna et al. 2024):

- **SVA**: singular/plural agreement (12,400 pairs)
- **Greater-than**: year comparison (1,000 pairs)
- **Gender bias**: occupation → he/she (986 pairs)
- **Capitals**: country → capital (190 pairs)
- **Hypernymy**: word → category (198 pairs)

Key question: does Structured pi-SAE's **conservatism** — only reporting high IIA when

Task	DAS	NL-DAS	Str. pi-SAE
$k = 1$			
SVA	0.00	1.00	0.00
Greater-th	0.03	1.00	0.00
Gender bias	0.88	1.00	0.54
Capitals	0.00	0.00	0.12
Hypernymy	0.03	0.20	0.50
$k = 4$			
SVA	0.02	1.00	0.00
Greater-th	0.31	1.00	0.00
Gender bias	1.00	1.00	0.95
Capitals	0.02	0.07	0.37
Hypernymy	0.08	0.43	0.47

Red: NL-DAS false positives (IIA ≈ 1 on tasks

Interpretation: Structured pi-SAE as a causal structure detector

Three regimes:

1. High-dimensional tasks (SVA,

greater-than):

- All methods except NL-DAS get ≈ 0
- NL-DAS gets ≈ 1.0 = **false positive**
- Structured pi-SAE correctly returns 0 — no low-dim structure

2. Genuinely low-dim (gender bias):

- Binary task (he/she), well-represented linearly
- DAS gets 0.88–1.0, Structured pi-SAE gets 0.54–0.95
- Structured pi-SAE recovers but needs $k \geq 2$

3. Nonlinear structure (capitals, hypernymy):

- DAS fails (≈ 0), NL-DAS also weak

Comparison with factorized DAS

Hard-mode IIA, layer 8, $k=1$, atomic-sweep-40:

Task	DAS	Fac. DAS
IOI	0.94	0.94
SVA	0.79	0.79
Gender	0.79	0.83
Capitals	0.69	0.69
Hypernymy	1.00	1.00

Factorized DAS matches optimal DAS exactly.

Structured pi-SAE complements: it **detects** nonlinear structure that DAS misses (capitals, hypernymy), but DAS has higher IIA on linearly-accessible variables (gender bias).

Together: structure detection (Structured pi-SAE) + subspace identification (factorized DAS).

11. Cross-task transfer

Cross-task transfer validates causal variables

A genuine causal variable should **transfer** — train on one task variant, test on another.

IOI cross-template transfer (GPT-2):

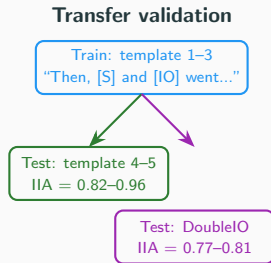
- Train VAE on templates 1–3
- Test on unseen templates 4–5
- IIA = 0.82–0.96 at $k = 4$

Extended IOI tasks^a:

- DoubleIO: IIA = 0.77–0.81
- TripleIO: IIA = 0.77–0.81
- Trained only on base IOI!

IOI subtask transfer^b (8 counterfactual types):

- DAS transfer ratio: 0.56



A lookup table cannot transfer like this

12. IOI is not Grassmannian

IOI is NOT Grassmannian: sheaf consistency analysis

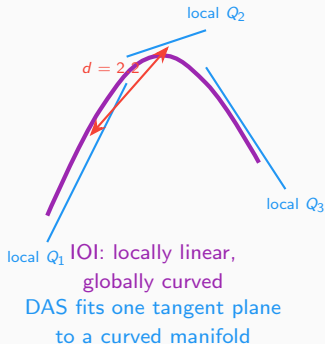
Even within GPT-2's IOI circuit, the causal variable is not globally linear.

We fit local DAS subspaces to different clusters of activations and measure consistency:

- Geodesic distance between local subspaces: ~ 2.2
- Cross-cluster IIA: ~ 0.21

If IOI were Grassmannian, local subspaces would agree (geodesic ≈ 0 , cross-IIA ≈ 1.0).

The subspace that represents the indirect object **rotates** across different activation clusters. Locally linear, globally nonlinear — like a curved manifold approximated by tangent planes.



From subspaces to interpretable factors

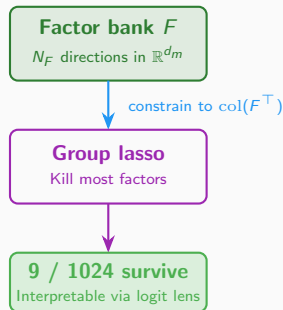
DAS finds a subspace. But which *directions* compose it?

If the model's weight matrices decompose as $W_i = S_i \cdot F$ through a shared **factor bank** F with sparse **selectors** S_i , then DAS can be constrained to search within the factor span:

Standard DAS: $U = \text{orth}(A)$, $A \in \mathbb{R}^{d_m \times k}$

Factorized DAS: $U = \text{orth}(F^\top A)$, $A \in \mathbb{R}^{N_F \times k}$

With a group lasso penalty ($\lambda \sum_i \|A_{i,:}\|_2$), most factors are killed. What survives is interpretable:



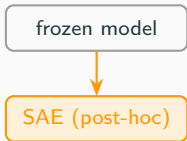
Standard DAS finds the subspace.
Factorized DAS finds the **factors** in it.

Task	Active	Total	Top factor decodes to
IOI	9	1024	wives, women, Ms
Gender bias	3	1024	she, her, herself
SVA	2	1024	are, is, aren't

Weight space vs. activation space

Both SAEs and sparse weight factorization produce dictionaries of directions in the residual stream. But they decompose different things:

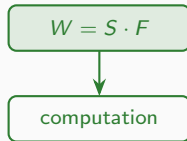
SAEs (activation space)^a



Decomposes what the model *outputs* at a layer.

Not guaranteed to capture what the model actually *uses*.

Factors (weight space)



Decomposes how the model *computes*:

$$W_i = S_i \cdot F.$$

Guaranteed to be used — every active factor is part of the computation.

^aCunningham et al., "Sparse Autoencoders Find Highly Interpretable Directions," ICLR 2024.

Part 5

Implications and practical advice

15. Four-level hierarchy

A hierarchy for causal validity: IIA alone is necessary but not sufficient. Faithfulness metrics and structural diagnostics provide increasingly strong evidence.

16. Continuous metrics

Binary IIA hides important differences. Continuous metrics (KL, logit diff) distinguish methods that all achieve $\text{IIA} \approx 1$.

17. Practical advice

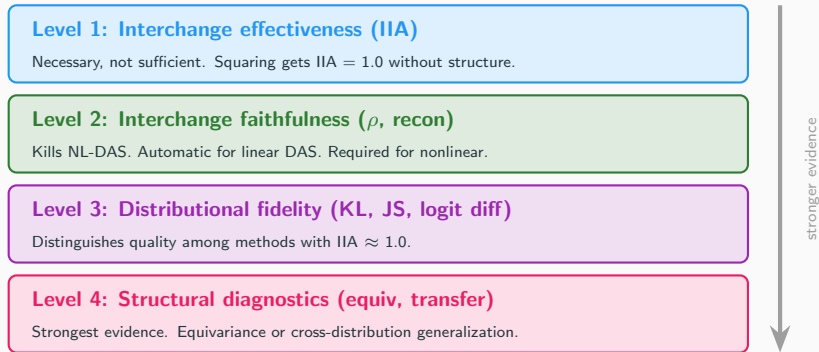
Five concrete recommendations for causal abstraction practitioners: report faithfulness, use continuous metrics, validate nonlinear methods, test transfer.

18. Conclusion

The linearity assumption is testable. We provide both the diagnostics and the method to handle nonlinear variables.

13. Four-level hierarchy

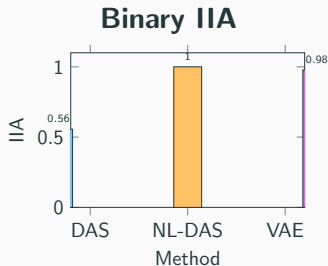
A four-level hierarchy for causal validity



Key asymmetry: Linear DAS automatically passes Level 2 (orthogonal projection is faithful by construction) but may fail Level 1 on nonlinear variables. NL-DAS easily passes Level 1 but fails Level 2.

14. Continuous metrics

Binary IIA hides important differences



NL-DAS “wins.” But it is a lookup table.

With faithfulness metrics

	DAS	NL-DAS	VAE
IIA	0.556	1.000	0.978
Diversity ρ	1.0	≈ 0	≈ 1
Transfer	Yes	No	Yes

VAE wins genuinely. NL-DAS is disqualified
by $\rho \approx 0$.

DAS is honest but limited by linearity.

15. Practical advice

Practical advice for causal abstraction

1. **Never report IIA alone.** $\text{IIA} = 1.0$ is compatible with genuine structure, memorized lookup tables, and degenerate encoder-decoders. Always report at least one faithfulness metric.
2. **Use continuous distributional metrics.** KL divergence and normalized logit difference distinguish among methods that all achieve $\text{IIA} \approx 1.0$. Binary IIA treats a 51% shift the same as 99%.
3. **Be suspicious of unconstrained nonlinear featurizers.** Any encoder-decoder trained end-to-end on interchange loss without reconstruction constraints can learn a lookup table. The ELBO (or equivalent) is necessary.
4. **Test cross-distribution generalization.** Evaluate on disjoint names, novel templates, or new task variants. A method that fails under distribution shift is exploiting surface regularities.
5. **Consider nonlinear methods when DAS IIA is low.** Low IIA does not mean no causal

Surprising positive and negative cases

Surprising positives

Cubic sum ($a^3 + b^3$): perfect equivariance (100%) despite cubing alone reaching only 50.9%. The boundary is not “group action vs. polynomial” but whether the binary structure provides sufficient additive symmetry.

Max ($\max(a, b)$): achieves 95.7% equivariance despite not being a group action. It is piecewise equivariant: when the argmax is stable under shift, max behaves like a shifted identity.

Surprising negatives

Affine ($2a + 3b + 5$): a *linear* function that *fails* to produce a Grassmannian variable! Under $a \mapsto a + g$, the output shifts by $2g$, not g . Equivariance requires the operation to be a group action, not merely linear.

Squaring: achieves $\text{IIA} = 1.0$ by $k = 4$ without grokking. High IIA is the illusion; low equivariance (31.4%) is the truth.

16. Conclusion

Summary

1. 14 operations $\rightarrow$ three sharp classes (Always / Stochastic / Never Grassmannian).
Boundary governed by grokking.

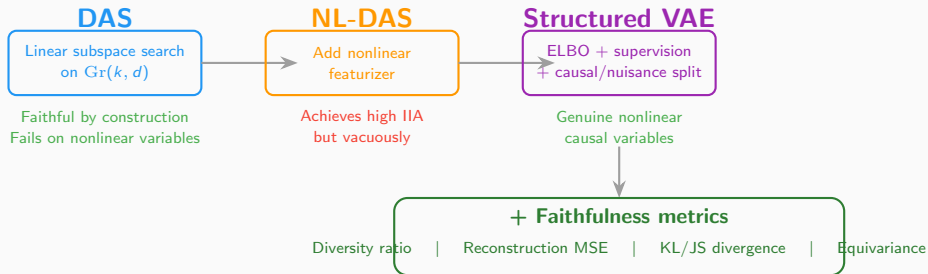
2. Linear DAS returns $\text{IIA} = 0$ on grokked addition. The variable lives on S^1 , not $\text{Gr}(k, d)$.
Category error.

3. Unconstrained NL-DAS achieves $\text{IIA} = 1.0$ vacuously (lookup table). Diversity ratio ρ exposes this.

4. Structured VAE recovers genuine nonlinear causal variables. ELBO prevents degeneracy.
Transfers across tasks.

The linearity assumption is **testable**. The natural fix (nonlinearity) introduces a new failure mode.
This paper provides the diagnostics to detect both problems and the method to avoid them.

The big picture



Every causal abstraction claim should be evaluated with the **four-level hierarchy**.

IIA is necessary but never sufficient.

Thank you

Code and experiments:

`github.com/elliottower/grokking-causal-geometry`

Elliot Tower · Ivan Vegner · N. Siddharth · Alex Doumas

Draft paper: *When Does Linear Causal Abstraction Work?*

Mapping the Boundary on the Grassmannian